

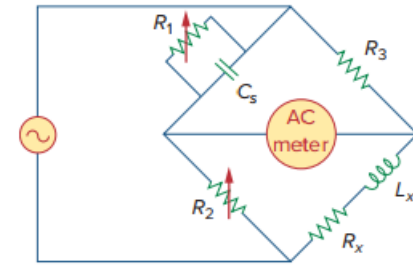
Problem

The ac bridge shown in Fig. 9.84 is known as a *Maxwell bridge* and is used for accurate measurement of inductance and resistance of a coil in terms of a standard capacitance C_s . Show that when the bridge is balanced,

$$L_x = R_2 R_3 C_s \quad \text{and} \quad R_x = \frac{R_2}{R_1} R_3$$

Find L_x and R_x for $R_1 = 40 \text{ k}\Omega$, $R_2 = 1.6 \text{ k}\Omega$, $R_3 = 4 \text{ k}\Omega$, and $C_s = 0.45 \mu\text{F}$.

Figure 9.84:



Step-by-step solution

Step 1 of 2

$$\text{Let } Z_1 = R_1 \parallel \frac{1}{j\omega C_s}, \quad Z_2 = R_2, \quad Z_3 = R_3,$$

$$\text{And } Z_x = R_x + j\omega L_x.$$

$$Z_1 = \frac{\frac{R_1}{j\omega C_s}}{R_1 + \frac{1}{j\omega C_s}} = \frac{R_1}{j\omega R_1 C_s + 1}$$

$$\text{Since, } Z_x = \frac{Z_3}{Z_1} Z_2,$$

$$R_x + j\omega L_x = R_2 R_3 \left(\frac{j\omega R_1 C_s + 1}{R_1} \right) = \frac{R_2 R_3}{R_1} (1 + j\omega R_1 C_s)$$

Step 2 of 2

Equating the real and imaginary components,

$$R_X = \frac{R_2 R_3}{R_1}$$

$$\omega L_X = \frac{R_2 R_3}{R_1} (\omega R_1 C_S) \quad \text{implies that}$$

$$L_X = R_2 R_3 C_S$$

Given that,

$$R_1 = 40 \text{ K}\Omega, \quad R_2 = 1.6 \text{ K}\Omega, \quad R_3 = 4 \text{ K}\Omega$$

And $C_S = 0.45 \mu F$

$$R_X = \frac{R_2 R_3}{R_1} = \frac{(1.6)(4)}{40} = 0.16 \text{ K}\Omega = 160 \Omega$$

$$L_X = R_2 R_3 C_S = (1.6)(4)(0.45) = 2.88 \text{ H}$$